

PROGRAMMING

CONTEST #2

CSE110

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Rules and Regulations

Team Building Rules

1. There will be eight teams. Eight horsemen/horsewomen will be selected by the instructor. The selection will be based on the class performance of the students. They will form teams according to their preference. Every student must participate.
2. Each team must have at least one person with knowledge on the following:
 - a. **Snakes and Ladders** Game
 - b. **Attack on Titan** Anime
 - c. Family Tree and Bootstrap Paradox of the **Dark** Series
 - d. **Set** Theories
 - e. Different Types of **Mathematical Series**
 - f. **Spotify** Playlists Modifier
 - g. **Minecraft** Mechanics

Contest Rules

1. The contest will be in a tournament mode.
2. In the lower bracket each team will have 3 matches. Wins=3 points, draw= 2 points, loss= 1 point
3. Top 3 teams of the lower bracket will go to the upper bracket.
4. The outputs must match exactly as shown in the question.
5. The code should work for every sample input/output combination.
6. One team should not aid any other team. If a team is struggling with something, that team can request help from the instructor.
7. One person from each team will submit the **.ipynb** file in the Google form.
8. All the eliminated participants can still compete to become the top 6 performers and get bonus.

Scoring Criteria

Each Solved Problem	7
Efficient and Clean Code	1
Presentation	2
Total	10

Perks and Benefits

Champion	+1.5 Point
First Runner Up	+1 Point
3 rd Place	+0.5 Point
All the participants eliminated in First Phase	-0.5 Points
Best 6 Performers of Top 3 Teams	+1.5 Points
Best 6 Performers of Bottom 3 Teams	+1 Point



Submission and Scorecard

Submission Form: <https://forms.gle/YiT1nTnHmu2acH1Q7> (Submit using G-suite Email)

Presentation Dates: 10, 15 (Online), 17, 22 and 24 August, 2022

Team Formation and Scorecard: <https://docs.google.com/spreadsheets/d/1vGK5-ZxskTzERceg-EiOLrjnyJ7D-QpIoVSw-yqoAUG/edit?usp=sharing>

Tournament Proceedings

Lower Bracket (10, 15, 17 August)

Round 1 [10 August]

1

6

Team 6

App #1

1

Team 1

2

2

Team 2

App #5

5

Team 5

3

3

Team 3

App #6

4

Team 4

Round 2 [15 August]

4

5

Team 5

App #1

3

Team 3

5

4

Team 4

App #5

6

Team 6

6

1

Team 1

App #6

2

Team 2

Round 3 [17 August]

7

4

Team 4

App #4

5

Team 5

8

3

Team 3

App #4

1

Team 1

9

6

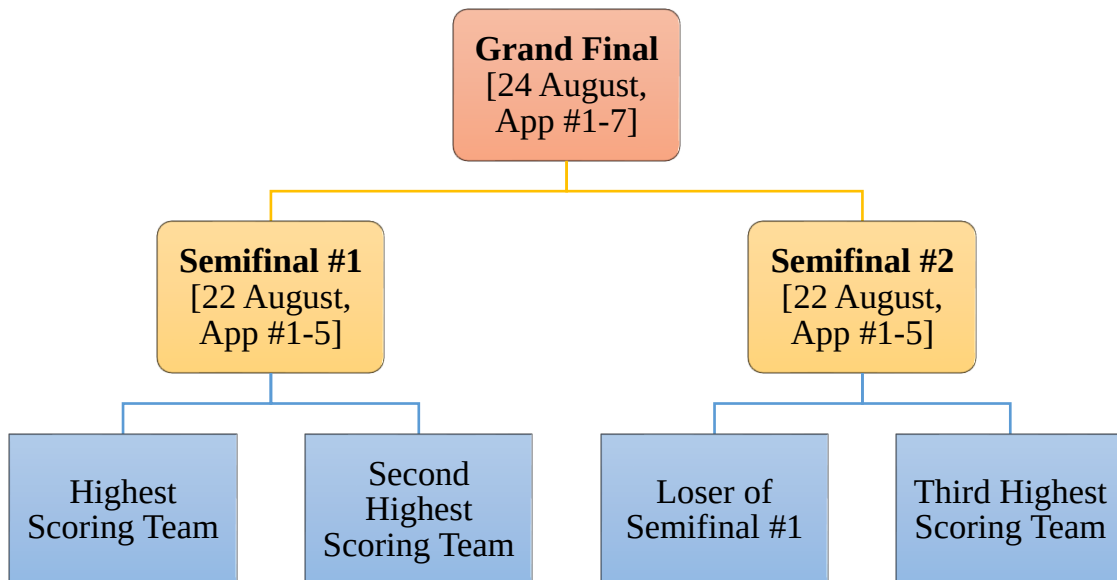
Team 6

App #4

2

Team 2

Upper Bracket (22 and 24 August)



App #1: Set Operations [Difficulty Level: As Easy as Maths]

Design a set operation application that asks the user which set operation the user wants to perform. Then take one/two **string** inputs from the user and convert these into list(s). After performing the appropriate set operation, print the resultant list.

- .sort() function is allowed
- Take conceptual help from: <https://www.cuemath.com/algebra/operations-on-sets/>

Sample Input #1: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 1 A: [10, 20, 30] B: [20, 30, 40, 50] Output: A $\cup$ B: [10, 20, 30, 40, 50]	Sample Input #2: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 2 A: [10, 20, 30] B: [20, 30, 40, 50] Output: A $\cap$ B : [20, 30]
Sample Input #3: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 1 A: [] B: [] Output: A $\cup$ B: []	Sample Input #4: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 2 A: [10, 20, 30] B: [] Output: A $\cap$ B: []
Sample Input #5: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 3 A: [10, 20, 30] Universal Set, U: [10, 20, 30, 40, 50] Output: A' : [40, 50]	Sample Input #6: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 3 A: [10, 20, 30, 70] Universal Set, U: [10, 20, 30, 40, 50] Output: Error: First Set Containing Elements that are not a Part of the Universal Set.
Sample Input #7: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 4 A: [10, 20, 30, 70, 80] B: [20, 30, 40, 50] Output: A – B: [70, 80]	Sample Input #8: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 4 A: [10, 20, 30, 70, 80] B: [40, 50] Output: A – B: [10, 20, 30, 70, 80]
Sample Input #7: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 5 A: [10, 20, 30] B: [30, 40] Output: A X B: [[10, 30], [10, 40], [20, 30], [20,40], [30, 30], [30, 40]]	Sample Input #8: Set Operation (1=Union, 2=Intersection, 3=Complement, 4=Difference, 5=Cross Product): 5 A: [10, 20, 30] B: [] Output: A X B: [[10, None], [20, None], [30, None]]

App #2: Minecraft Crafting Table [Difficulty Level: As Easy as Mining Bedrock]

Imagine due to multiple realities collapsing in on each other, we ended up in an entirely different dimension, where everything is made of Minecraft blocks. Now in order to survive the hordes of zombies and skeletons, you have to quickly design a program that provides you with the recipe of crafting a few necessary items. The recipe includes all the ingredients needed to craft an object of your choosing, the sub-ingredients needed to craft the ingredients and so on. The process will continue until there are no more sub-ingredient.

Copy the following dictionary directly in your code. If a specific item is not found within this dictionary, simply print "Recipe not found". Otherwise, print a dictionary containing all the ingredients (including quantity) necessary to craft that particular item.

```
craft= { "redstone torch": {"torch": 1, "redstone": 1}, "iron sword": {"iron ingot": 2, "stick": 1}, "torch": {"stick": 1, "coal": 1}, "stick": {"planks": 1}, "boat": {"planks": 5}, "planks": {"oak log": 1}, "bed": {"wool": 3, "planks": 3}, "beacon": {"nether star": 1, "glass": 5, "obsidian": 3}, "furnace": {"cobblestone": 8}, "blast furnace": {"furnace": 1, "iron ingot": 5, "smooth stone": 3} }
```

Sample Input #1: redstone torch Output: { "torch": 1, "redstone": 1, "stick": 1, "coal": 1, "planks": 1, "oak log": 1 }	Sample Input #2: iron sword Output: { "iron ingot": 2, "stick": 1, "planks": 1, "oak log": 1 }
Sample Input #3: grindstone Output: Recipe not found	Sample Input #4: blast furnace Output: { "furnace": 1, "iron ingot": 5, "smooth stone": 3, "cobblestone": 8 }

**POV: SHE DIED AND
RESPAWNED IN YOUR FRIENDS BED**



App #3: Attack on Titan [Difficulty Level: Fighting Levi Ackerman]

Throughout history 9 legendary titans have existed. These titans are: Founding, Attack, Armored, Colossal, Female, Cart, Jaw, Beast and Warhammer. A titan shifter gets access to another titan shifter's power through devouring them.

Within the last 200 years, there have been many battles among the titans and the titan powers have switched hands many times. Your job is to find out the latest wielders of the titans.

You have two dictionaries below which you must copy directly onto your code. The first dictionary shows who eats whom and the second dictionary shows the original wielders of each of the titans.

***Do not worry about being spoiled :P This problem contains fake spoilers. ***

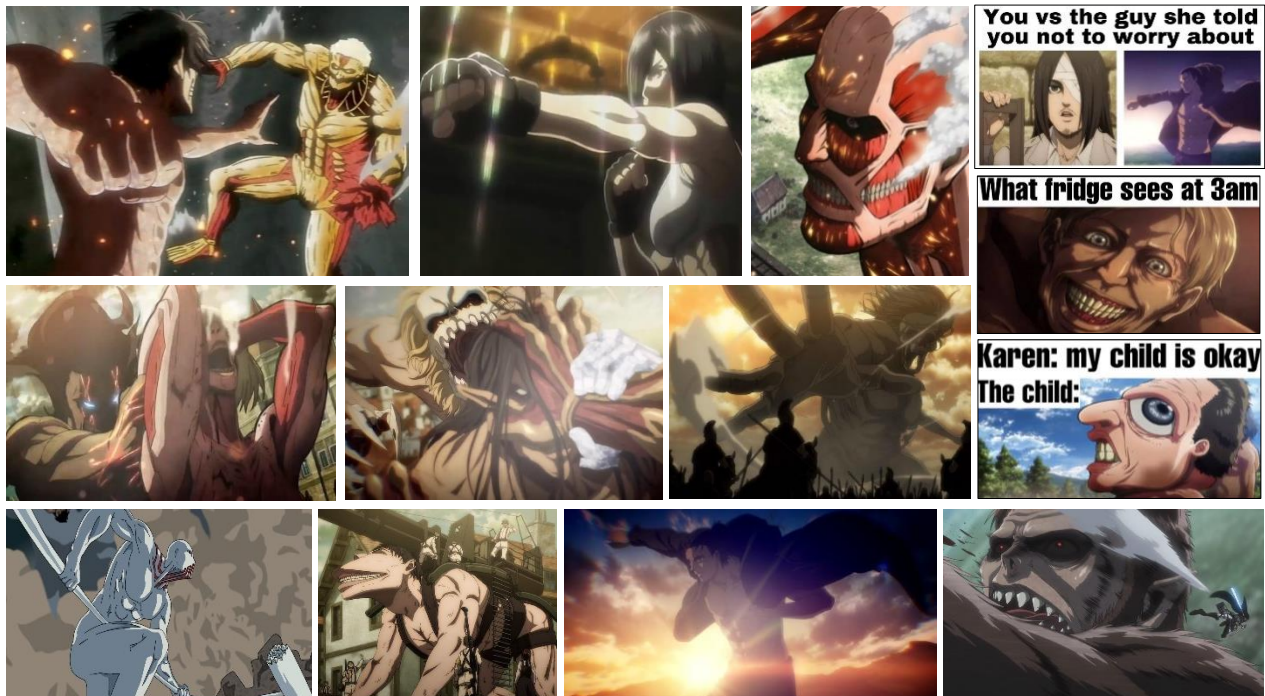
Given Dictionaries:

```
titan_eaters= {"Eren Yeager": ["Grisha Yeager", "Lara Tybur"], "Grisha Yeager": ["Eren Kruger", "Frieda Reiss"], "Porco": ["Marcel"], "Marcel": ["Ymir"], "Falco": ["Porco"], "Armin": ["Pieck"], "Levi Ackerman": ["Zeke Yeager"], "Mikasa Ackerman": ["Levi", "Annie"], "Commander Erwin": ["Berethold"]}
```

```
original_titans= {"Founder": "Frieda Reiss", "Attack": "Eren Kruger", "Armored": "Reiner", "Colossal": "Berethold", "Female": "Annie", "Jaw": "Ymir", "Cart": "Pieck", "Warhammer": "Lara Tybur", "Beast": "Zeke Yeager"}
```

Output:

```
{"Eren Yeager": ["Founder", "Attack", "Warhammer"], "Falco": ["Jaw"], "Mikasa Ackerman": ["Beast", "Female"], "Commander Erwin": ["Colossal"], "Reiner": ["Armored"], "Armin": ["Cart"]}
```



App #4: Dark Bootstrap Paradox [Difficulty Level: Finding the Origin of the Paradox]

Bootstrap Paradox

Bootstrap Paradox occurs when an object, information or a person travels back in time and gets trapped in an infinite cause and effect loop and no longer has a point of origin.

Imagine you have been recruited by Future Adam and Claudia of the famous series *Dark*. Now, you have been tasked to list and investigate on all the paradoxes. You have to design a program that should be able to detect the number of paradoxes present in the events and print each of the paradox. Copy the events string directly into your code.



events= "Michael gets lost in the woods. Ulrich goes to find Michael. Both travel to the past but not in the same timeline. Ulrich gets put into the prison. Michael is Jonas's father. Michael committed suicide by hanging himself. Jonas travels back in time to save his father. He finds his dad and tells him about his death. His dad now understood what he had to do. Michael committed suicide by hanging himself. The entire family of Tannahauss was killed in a car accident. This drove him crazy. Tannahauss built a time machine. He also wrote a book on time travel. Claudia finds the book on time travel. Future Claudia gave the present Claudia and time machine. The present Claudia travels back in time and gave the time travelling book to Tannahauss. Tannahauss took ideas from the book he was supposed to write in future. Tannahauss built a time machine. A cult like faction keeps on abducting kids from Winden and experiments on them. None of the abducted kids are ever found again. Charlotte gave birth to Elizabeth. The apocalypse takes place and Charlotte dies. However, Elizabeth somehow survives and gets sent to the past. Elizabeth meets a guy and they have a daughter named Charlotte. After growing up Charlotte gets married. Charlotte gave birth to Elizabeth. Thus the cycle continues. After the apocalypse, only a handful of people survived. Elizabeth was one of them. She became the leader of a savage group. Jonas and Martha love each other. Martha is killed by Adam. Adam is from the future. Jonas travels through time to stop Adam, only to find out that he is the one who eventually becomes Adam. Adam is the only one capable of breaking the endless cycle. For Jonas to become Adam, it is necessary for Martha to die. Without her dying, Jonas would not become Adam and the cycle would never be broken. So, Adam goes back in time. Martha is killed by Adam. After Martha's death Jonas travels into another world."

Output:

4 Paradoxes

Michael committed suicide by hanging himself. Jonas travels back in time to save his father. He finds his dad and tells him about his death. His dad now understood what he had to do. Michael committed suicide by hanging himself.

Tannahauss built a time machine. He also wrote a book on time travel. Claudia finds the book on time travel. Future Claudia gave the present Claudia and time machine. The present Claudia travels back in time and gave the time travelling book to Tannahauss. Tannahauss took ideas from the book he was supposed to write in future. Tannahauss built a time machine.

Charlotte gave birth to Elizabeth. The apocalypse takes place and Charlotte dies. However, Elizabeth somehow survives and gets sent to the past. Elizabeth meets a guy and they have a daughter named Charlotte. After growing up Charlotte gets married. Charlotte gave birth to Elizabeth.

Martha is killed by Adam. Adam is from the future. Jonas travels through time to stop Adam, only to find out that he is the one who eventually becomes Adam. Adam is the only one capable of breaking the endless cycle. For Jonas to become Adam, it is necessary for Martha to die. Without her dying, Jonas would not become Adam and the cycle would never be broken. So, Adam goes back in time. Martha is killed by Adam.

App #5: Snakes and Ladders

[Difficulty Level: Walk in the Park]

Imagine you are building the snakes and ladders game by yourself for two players. The following are the rules of the game:

- Both players will start from box 0 and only get to box 1 after scoring a 1. Otherwise, they will stay in box 0.
- The one reaching box 30 first is the winner
- Any player who reaches the head of a snake will be eaten and sent to the tail of that snake.
- Any player who reaches the bottom of the ladder gets elevated to the top of that ladder.
- If a player gets to box 29, he/she must score 1 to win.

The following two dictionaries are the snakes and ladders positions respectfully. In the first dictionary, the keys are the heads and the values are the tails of the snakes. In the second dictionary, the keys are the bottom of the ladders and the values are the top of the ladders. You can copy these two dictionaries directly onto your code.

Snakes= {15: 5, 19: 12, 25: 16, 27: 21}

Ladders= {9: 14, 13: 18, 17: 26}

You have to find the winner and print the position of the two players after each turn. You also have to show whenever a player gets a ladder or gets eaten by a snake. You have to continuously take P1 and P2 inputs until a winner is decided.

Sample Input/Output 1:	Sample Input/Output 2:
P1: 1 P2: 4 P1 in box 1 and P2 in box 0 P1: 5 P2: 1 P1 in box 6 and P2 in box 1 P1: 3 P1 got a ladder! P2: 4 P1 in box 14 and P2 in box 5 P1: 5 P1 got eaten by a snake! P2: 6 P1 in box 12 and P2 in box 11 P1: 5 P1 got a ladder! P2: 4 P2 got eaten by a snake! P1 in box 26 and P2 in box 5 P1: 3 P2: 6 P1 in box 29 and P2 in box 11 P1: 3 P2: 6 P2 got a ladder! P1 in box 29 and P2 in box 26 P1: 1 P1 Wins!	P1: 4 P2: 6 P1 in box 0 and P2 in box 0 P1: 1 P2: 1 P1 in box 1 and P2 in box 1 P1: 4 P2: 5 P1 in box 5 and P2 in box 6 P1: 4 P1 got a ladder! P2: 6 P1 in box 14 and P2 in box 12 P1: 5 P1 got eaten by a snake! P2: 5 P2 got a ladder! P1 in box 12 and P2 in box 26 P1: 5 P1 got a ladder! P2: 6 P1 in box 26 and P2 in box 26 P1: 3 P2: 4 P2 Wins!

App #6: Mathematical Series Builder

[Difficulty Level: As Easy as Maths]

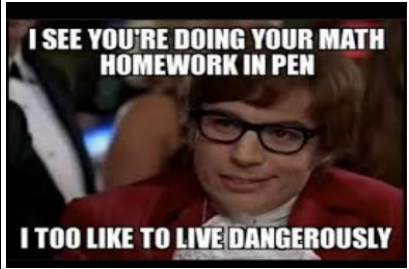
Imagine you got frustrated with CSE and now looking to shift into MNS. Some of your friends encouraged you to do a double major in both CSE and MNS. You being a stubborn child, decided to have a go at it. Now, you want to build a program that blends both CSE and MNS altogether.

Design an application that asks which series you want to build and provides the sum of the series depending on the **extent** (n) of that particular series. Value of n must be greater than 0. Assume all the inputs are integers.

Your application should be able to build the following types of series:

1. Common Difference Series [a, (a+d), (a+2d), , (a+(n-1) d)]
2. Power Series [1^x , 2^x , 3^x , ..., n^x]
3. Fibonacci Series [0, 1, 1, 2, 3, 5, 6....]
4. Reversal Series [1, -2, 3, -4, ..., (+ or -) n]

Therefore, the valid choices for selecting series range in between 1 to 4.

Sample Input #1: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 1 Value of n: 5 Value of a: 1 Value of d: 3 Output: Series: 1, 4, 7, 10, 13 Sum of the Series: 35	Sample Input 2: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 2 Value of n: 5 Value of x: 2 Output: Series: 1^2 , 2^2 , 3^2 , 4^2 , 5^2 Sum of the Series: 55	Sample Input 3: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 2 Value of n: 4 Value of x: 3 Output: Series: 1^3 , 2^3 , 3^3 , 4^3 Sum of the Series: 100
Sample Input 4: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 3 Value of n: 7 Output: Series: 0, 1, 1, 2, 3, 5, 8 Sum of the Series: 20	Sample Input 5: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 4 Value of n: 6 Output: Series: 1, -2, 3, -4, 5, -6 Sum of the Series: -3	Sample Input 5: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 4 Value of n: 7 Output: Series: 1, -2, 3, -4, 5, -6, 7 Sum of the Series: 4
Sample Input 7: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 2 Value of n: -1 Output: Invalid Input	Sample Input 8: Which Series to Build? (1= Common Difference, 2= Power, 3= Fibonacci, 4= Reversal): 5 Output: Invalid Choice	 I SEE YOU'RE DOING YOUR MATH HOMEWORK IN PEN I TOO LIKE TO LIVE DANGEROUSLY

App #7: Spotify Playlist Modifier

[Difficulty Level: In the end it doesn't even matter]

Imagine your Spotify subscription just ended and you are thinking about renewing your subscription. However, you are not satisfied with some of the features of Spotify. So, you took it upon yourself to design a playlist modifier for Spotify which will add, delete, search, or sort songs in a playlist; create new playlists, blend multiple playlists together or delete playlists.

You should have a dictionary that holds all your playlists. Each playlist will have a list of its own that holds its corresponding songs. Your program should have the following functions:

1. create_playlist (name)
2. add_songs (playlist, song)
3. delete_songs (playlist, song)
4. search_songs (playlist, song)
5. sort_songs (playlist)
6. blend_playlists (playlist_1, playlist_2, new_name)
7. delete_playlist (playlist)
8. show_playlist (playlist)
9. show_all_playlists ()



Function Calls #1

```
create_playlist ("Chill Hour")
add_songs ("Chill Hour", "Hello")
add_songs ("Chill Hour", "Beautiful Mistakes")
add_songs ("Chill Hour", "Echoes in Your Attic")
show_playlist ("Chill Hour")
search_songs ("Chill Hour", "Beautiful Mistakes")
delete_songs ("Chill Hour", "Beautiful Mistakes")
search_songs ("Chill Hour", "Beautiful Mistakes")
delete_songs ("Chill Hour", "In The End")
sort_songs ("Chill Hour")
create_playlist ("Come vibe with me")
add_songs ("Come vibe with me", "Apocalypse")
show_all_playlists ()
blend_playlists ("Chill Hour", "Come vibe with me", "Colab")
show_playlist ("Colab")
delete_playlist ("Chill Hour")
show_playlist ("Chill Hour")
```

Output

```
New Playlist Created: Chill Hour
Song Hello added to Chill Hour
Song Beautiful Mistakes added to Chill Hour
Song Echoes in Your Attic added to Chill Hour
Chill Hour: ["Hello", "Beautiful Mistakes", "Echoes in Your Attic"]
Beautiful Mistakes found at position 2 in Chill Hour
Beautiful Mistakes removed from Chill Hour
Beautiful Mistakes not found in Chill Hour
No song named In The End in Chill Hour
Chill Hour playlist sorted alphabetically
New Playlist Created: Come vibe with me
Song Apocalypse added to Come vibe with me
{"Chill Hour": ["Beautiful Mistakes", "Echoes in Your Attic", "Hello"], "Come
vibe with me": ["Apocalypse"]}
Chill Hour and Come vibe with me blended into Colab
Colab: ["Beautiful Mistakes", "Echoes in Your Attic", "Hello", "Apocalypse"]
Playlist: Chill Hour deleted
No playlist named Chill Hour found
```

Function Calls #2

```
create_playlist ("Twisted Bliss")
add_songs ("Twisted Bliss", "Middle of the Night")
add_songs ("Twisted Bliss", "After Hours")
add_songs ("Twisted Bliss", "Levitating")
show_playlist ("Twisted Bliss")
search_songs ("Twisted Bliss", "Levitating")
delete_songs ("Twisted Bliss", "Awake")
search_songs ("Twisted Bliss", "Awake")
delete_songs ("Twisted Bliss", "Letters from the Sky")
sort_songs ("Twisted Bliss")
create_playlist ("JoJo")
add_songs ("JoJo", "The Pillar Men Theme")
show_all_playlists ()
blend_playlists ("JoJo", "Twisted Bliss", "Colab2")
show_playlist ("Colab2")
delete_playlist ("Love or Hate")
show_playlist ("JoJo")
```

Output

```
New Playlist Created: Twisted Bliss
Song Middle of the Night added to Twisted Bliss
Song After Hours added to Twisted Bliss
Song Levitating added to Twisted Bliss
Twisted Bliss: ["Middle of the Night", "After Hours", "Levitating"]
Levitating found at position 3 in Twisted Bliss
No song named Awake in Twisted Bliss
Awake not found in Twisted Bliss
No song named Letters from the Sky in Twisted Bliss
Twisted Bliss playlist sorted alphabetically
New Playlist Created: JoJo
Song The Pillar Men Theme added to JoJo
{"Twisted Bliss": ["After Hours", "Levitating", "Middle of the Night"], "JoJo": ["The
Pillar Men Theme"]}
Twisted Bliss and JoJo blended into Colab2
Colab: ["The Pillar Men Theme", "After Hours", "Levitating", "Middle of the Night"]
No playlist named Love or Hate found
"JoJo": ["The Pillar Men Theme"]
```